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Effect of Lanthanum on Zr–Co/γ-Al_2O_3 Catalysts for Fischer–Tropsch Synthesis

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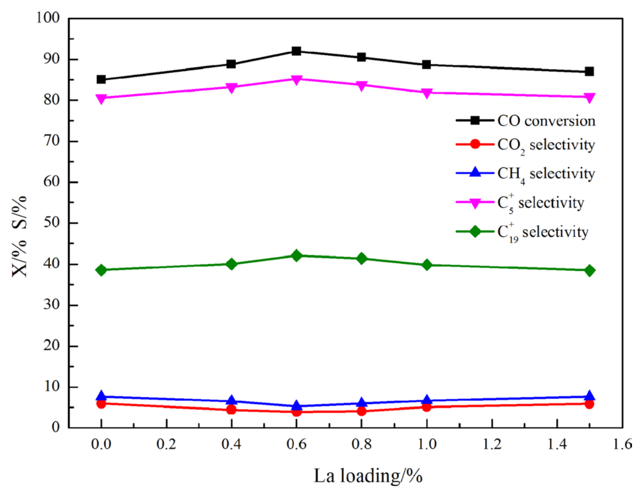
Abstract
Lanthanum promoted Zr–Co/γ-Al_2O_3 catalysts, with La loading from 0 to 1.5 wt%, were prepared by the incipient wetness method for Fischer–Tropsch. The impact of La content on the performance of the catalysts was studied by N_2 physisorption, XRD, H_2-TPR, H_2-TPD, CO-TPD and XPS techniques. It is found that the addition of La can affect the interaction between cobalt and support, enhance the cobalt reducibility and increase cobalt active sites, improving catalysts activity and selectivity to heavy hydrocarbons and restraining methane and CO_2 selectivity. Comparing with other catalysts, a higher activity with a 91.97% CO conversion and 85.23% C_5^+ selectivity at 220 °C, 2 MPa, and a WHSV of 1000 mL g^{-1} h^{-1} was achieved by the 0.6La–Zr–Co catalyst.

Graphical Abstract



Keywords Fischer–Tropsch synthesis · Zr–Co/γ-Al_2O_3 catalyst · Lanthanum

1 Introduction

As a mean of liquefaction of natural gas to a large variety of products such as paraffin, olefins, alcohols and aldehydes, Fischer–Tropsch (F–T) synthesis has gained the attention in academia and industry as an alternative to the use of limited resources of crude oil recently [1, 2]. Cobalt-based and iron-based catalyst have been widely used in F–T synthesis. Iron-based catalysts have better selectivity for the

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olefin and gasoline, but they produce more CO_2 and have lower durability [3, 4]. Compared to iron-based catalysts, cobalt-based catalysts have higher activity, higher long-chain paraffin and C_5^+ selectivity, relatively lower activity for the water–gas shift reaction [5]. Thus, cobalt-based catalysts are very effective for obtaining heavy hydrocarbon in F–T synthesis. In previous studies [6–8], it has been observed that supports could not only largely determine the number of active sites after reduction, but also strongly influence the percentage of the cobalt oxide species. Al_2O_3 is usually adopted as the support to prepare cobalt based catalysts for the excellent texture although the catalyst exhibits limited reducibility due to the strong interaction between the cobalt and the alumina support [9].

Many studies have demonstrated that the addition of promoters (e.g., Pt [10], Ru [11], ZrO_2 [12, 13], La_2O_3 [2, 14, 15]) could increase the reducibility of the cobalt-based catalysts, improving the activity and selectivity to heavy hydrocarbon. The effect of Zr promoter on the catalytic behavior of cobalt-based catalysts for F–T synthesis was investigated [16, 17]. Xiong et al. [18] reported that Zr addition could improve the activity and C_5^+ selectivity of Co/γ-Al_2O_3 catalyst and inhibit the formation $CoAl_2O_4$. Li et al. [19] explained that the ZrO_2 modification of γ-Al_2O_3 support could weaken the interaction between support and cobalt. Thus, it could be summarized that with the addition of zirconium, the formation of $CoAl_2O_4$ phase on the catalysts would be suppressed and Co metal active sites could increase. In some previous reports, lanthanum could also be utilized as a promoter for cobalt based F–T synthesis catalysts because of its beneficial effects, such as enhancing cobalt reducibility and cobalt dispersion [2, 20]. George et al. [2] studied that La^{3+} promotion of Co/SiO_2 could moderate strong Co-support interactions and enhance the reducibility of the Co oxide. Vada et al. [15] proposed that with increasing La_2O_3 loading, the hydrogen chemisorption on Co/Al_2O_3 was significantly blocked, resulting in increasing selectivity for heavier hydrocarbons. However, some controversies about the effect of lanthanum especially with the higher content of La-doped are still existing. Xiong et al. [21] studied that the strong interaction between Co and La would bring about the decreased reducibility of the catalysts, especially with high La loadings. Thus, the effect of lanthanum promoter still needs to be further researched and there are few reports concerning the interaction between zirconium and lanthanum for alumina supporting cobalt-based catalysts in F–T synthesis.

This paper focuses on studying the role of lanthanum as a promoter over Zr modification of Co/γ-Al_2O_3 catalysts for F–T synthesis. A series of Zr–Co/γ-Al_2O_3 catalysts modified by La were prepared with a range of La loading. The prepared catalysts were characterized by N_2 adsorption, X-ray diffraction (XRD), temperature programmed reduction (TPR), temperature programmed desorption of adsorbed CO (CO-TPD), hydrogen temperature programmed desorption (H_2-TPD), and X-ray photoelectron spectra (XPS) and were performed in a fixed bed reactor.

2 Experimental

2.1 Catalyst Preparation

The catalysts were prepared by the incipient wetness impregnation of the γ-Al_2O_3 support with aqueous lanthanum nitrate ($La(NO_3)_3·6H_2O$), zirconium nitrate ($Zr(NO_3)_4·5H_2O$) and cobalt nitrate ($Co(NO_3)_2·6H_2O$) solutions. The support γ-Al_2O_3 was first calcined at 450 °C for 6 h with a ramp to a temperature at 2 °C min^{-1} before impregnation. For Co/γ-Al_2O_3 catalyst, cobalt nitrate was dissolved in deionized water and directly impregnated into the support using incipient wetness to give a final catalyst with 25 wt% cobalt. The catalyst was dried at 110 °C for 12 h and calcined at 450 °C for 6 h.

For the Zr modified catalyst, zirconium was added to the Al_2O_3 support prior to the addition of Co. The zirconium nitrate was dissolved in an appropriate volume of deionized water and impregnated onto the γ-Al_2O_3 support with 4 wt% zirconium. After being dried and calcined by the same condition with the Co/γ-Al_2O_3 catalyst, cobalt nitrate was used to impregnate the Zr-modified supports to produce a calcined catalyst with 25 wt% cobalt.

La-doped Zr–Co/γ-Al_2O_3 catalysts, with constant Co content of 25 wt%, Zr loading of 4 wt% and La content from 0 to 1.5 wt%, were prepared by the previous method, except the first step that the lanthanum nitrate and the zirconium nitrate were dissolved in an appropriate volume of deionized water and impregnated onto the γ-Al_2O_3 support at the simultaneously. Then the following used the same procedure as given in the previous section. The prepared catalysts were represented as xLa–Zr–Co, where x indicated the lanthanum weight content (wt%, x = 0, 0.4, 0.6, 0.8, 1.0, and 1.5).

2.2 Catalyst Characterization

The specific surface areas, pore size distribution and pore volumes of catalysts were obtained by nitrogen adsorption at −196 °C using a Micromeritics ASAP 2020 (USA). Prior to the measurements, the sample was degassed for 6h at 350 °C under vacuum.

The X-ray diffraction patterns were recorded on a Rigaku D/MAX 2550 VB/PC X-ray diffractometer with Cu Kα radiation at 40 kV and 100 mA. The data were collected in the 2θ range of 5°–50° with a step size of 0.02°. The average Co_3O_4 crystallite size was calculated by the Shcherrer equation [22] from the most intense Co_3O_4 peak (2θ = 36.8°).

$$d = \frac{k\lambda}{B\cos\theta}\frac{180°}{\pi} \quad (1)$$

where d is the mean crystallite diameter, k (0.89) is the Scherrer constant, λ is the X-ray wavelength (1.54056), and B is the full width half maximum (FWHM) of Co_3O_4 diffraction peak.

The reduction behaviors of the prepared catalysts were analyzed by temperature programmed reduction. TPR experiment was carried out by AutoChem II 2920 (Micromeritics Co., USA). The catalyst was stacked in a quartz reactor with the content of 200 mg, fitted with a thermocouple for continuous temperature measurement. Prior to the temperature programmed reduction measurement, the sample was flushed under a flow of Ar (99%, 30 mL min^{-1}) at 300 °C for 2 h in order to remove water or impurities and then cooled down to room temperature. Then 50 mL min^{-1} of the reducing gas (10% H_2 in Ar balanced) was switched on and the temperature was increased at a ramp rate 10 °C min^{-1} up to 800 °C. The H_2 consumption (TCD) was recorded automatically.

Temperature programmed desorption of adsorbed CO was carried out in a flow apparatus of America Micromeritics Autochem 2910. The catalyst sample (ca.100 mg) was placed in a quartz reactor and reduced by flowing H_2 at 400 °C for 1 h, then purged by He at the same temperature for 0.5 h. When it was cooled to room temperature, 5% CO–He was introduced in pulse mode for adsorption. After CO adsorption saturation was achieved, the catalyst was swept with He for 30 min. Finally, the temperature was elevated from room temperature to 800 °C at a heating rate of 10 °C in a He flow at 20 mL min^{-1}. The product [m/z = 15(CH_4), 18(H_2O), 28(CO) and 44(CO_2)] was simultaneously detected with a Balsers OmniStar 300.

Hydrogen temperature programmed desorption was also carried out in a U-tube quartz reactor with the same instrument as the TPR. The sample weight was about 0.200 g. The catalyst was reduced at 350 °C for 2 h using a flow of high purity hydrogen and then cooled to 50 °C under the hydrogen stream. The sample was held at 50 °C for 1 h for H_2 adsorption. And then increasing the temperature slowly to 800 °C under the flowing Ar to desorb the remaining chemisorbed hydrogen and the TCD began to record the signal till the signal returned to the baseline.

The procedure of H_2-pulse chemisorption was similar with the H_2-TPD generally. The samples were reduced under high purity hydrogen at 350 °C for 2 h and then cooled to 50 °C under the hydrogen stream. Calibrated pulses of H_2 were injected into the system repeatedly until no further H_2 consumption was noted, and the time between pulses was 4 min.

The binding energy of Co was conducted on the X-ray photoelectron spectra measurement by an ESCALAB 250xi system (Thermo Fisher Scientific, UK) with Al Kα radiation (hv = 1486.6 eV), and calibrated by referencing to the C 1s peak form the contaminated carbon that was assumed to have a binding energy of 284.6 eV.

2.3 Catalyst Activity Testing

F–T synthesis reaction was carried out in a fixed-bed, stainless flow micro-reactor (6 mm I.D.). 3 g catalysts were stacked in the middle of the reactor while the quartz sand was added in both ends of the uniform temperature zone to keep the catalyst in the thermostatic area. The reactor temperature was increased from ambient to 120 °C in 60 min (hold 60 min) in a weight hourly space velocity (WHSV) of 1000 mL g^{-1} h^{-1} of H_2, then, increased to 430 °C in 6 h and held at that temperature for 12 h. Subsequently, the reactor was cooled down to 100 °C by flowing N_2. After reduction, the syngas (H_2:CO = 2.0) was introduced to the reactor and the pressure was increased to 2.0 MPa. The reactor temperature was raised to 200 °C in 3 h, then, increased to 220 °C in 2 h and the reaction was carried out at 220 °C in 1000 mL g^{-1} h^{-1}. The carbon balance was 100 ± 5%. After Fischer–Tropsch reaction, the wax and water mixture were collected in a hot trap, and the oil plus water was collected in a cold trap. The outlet gases CO, H_2, CH_4, C_2H_6, etc., were analyzed by an online GC Agilent 7890 A with a thermal conductivity detector (TCD). The wax dissolved in CS_2 and the oil were detected off-line by GC Agilent 7890 A with a flame ionization detector (FID). The carbon monoxide conversion (X_{CO}%) and the hydrocarbon selectivity were calculated based on the gas product analysis and the gas flow measuring results at steady state. Here, we define:

CO conversion:

$$X_{CO}(\%) = \frac{N_{CO,in} - N_{CO,out}}{N_{CO,in}} \times 100\% \quad (2)$$

C_1, $C_{2\sim4}$, C_5^+, C_{19}^+ selectivity:

$$S_{C_n}(\%) = \frac{nN_{C_n,out}}{N_{CO,in} - N_{CO,out}} \times 100\% \quad (3)$$

$$S_{CO_2}(\%) = \frac{N_{CO_2,out}}{N_{CO,in} - N_{CO,out}} \times 100\% \quad (4)$$

where $N_{i,\ in}$ and $N_{i,\ out}$ are the inlet and outlet mole flow rate (mol h^{-1}) of species i (i = CO, CO_2, C_1, $C_{2\sim4}$, C_5^+, C_{19}^+), respectively; n is carbon number of production (C_1, $C_{2\sim4}$, C_5^+, C_{19}^+).

3 Results and Discussion

3.1 BET Surface Area

The texture properties of the La-doped Co/γ-Al_2O_3 catalysts modified by Zr with different La loadings were characterized by using the N_2 adsorption. The BET surface areas (S_{BET}), pore volume (V_P), average pore diameter (D_P) of the supports and the catalysts are summarized in Tables 1 and 2. The pore size distributions of the catalysts have been shown in Fig. 1a, b. With increasing loading of La, the BET surface area of catalysts decreases while the average pore diameter increases and the average pore volume remains relatively unchanged, which are possibly attributed to the existence of La species on the surface of the catalyst leading to smaller surface areas, and some of the pores of γ-Al_2O_3 support are blocked.

3.2 X-Ray Diffraction

The XRD patterns for calcined catalysts and reduced catalysts are given in Fig. 2a, b. From the Fig. 2a, the apparent diffraction peaks at 31.3°, 36.8°, 45°, 59.4°and 65.4° derive from Co_3O_4 spinel phase. No evident diffraction peaks of any Zr or La compound are detected due to the low contents and high dispersion of Zr and La [23, 24]. The diffraction peak at 36.8° has been identified as the cubic spinel phase of Co_3O_4, and the average crystalline size of Co_3O_4 are calculated according to Scherrer's equation. As shown in Table 2, the average particle size of Co_3O_4 decreases with the increase of appropriate La content, indicating La has an impact on the cobalt dispersion and the aggregation of cobalt species. It is difficult to determine the $CoAl_2O_4$ phases based on the XRD characteristic of calcined catalysts alone because both Co_3O_4 and $CoAl_2O_4$ have cubic spinel structure with almost identical diffraction peak position [25]. So, the reduced catalysts XRD result can be used to observe the diffraction peak of $CoAl_2O_4$. From Fig. 2b, it is found that the intensity of $CoAl_2O_4$ diffraction peaks are slightly decreased with the increase of proper La, indicating that the addition of La can inhibit forming $CoAl_2O_4$ to some extent. The results have good agreement with H_2-TPR and XPS data. And then the Co metal particle size calculated by Scherrer's equation is shown in Table 2. The results show that the average size of cobalt particles decrease with the loading of 0.6 wt% La. It can be revealed that the existence of La hinders the aggregation of cobalt species, which can also be explained by H_2-TPD results. However, the particle size of Co_3O_4 and the $CoAl_2O_4$ diffraction peaks increase with the excessive La loading, which should be attributed to that large content of La can block some pores of catalysts, hinder the cobalt dispersion, and make its inhibiting effect on the interaction between cobalt and support decrease.

Table 1 Surface area, pore volume, average pore size for the supports

Catalyst samples	S_{BET} ($m^2\ g^{-1}$)	V_P ($cm^3\ g^{-1}$)[a]	D_P (nm)[b]
γ-Al_2O_3	221	0.46	5.75
0La–Zr	218	0.43	5.84
0.4La–Zr	215	0.42	5.88
0.6La–Zr	212	0.43	6.02
0.8La–Zr	212	0.43	6.06
1.0La–Zr	211	0.44	6.09
1.5La–Zr	210	0.42	6.12

[a]BJH desorption pore volume

[b]BJH desorption average pore size

3.3 H_2-Temperature Programmed Reduction

The TPR profiles of catalysts are showed in Fig. 3. There are two major reduction peaks exhibited for all samples and no obvious reduction peaks of Zr and La shown in results. The first lower-temperature peaks are attributed to the reduction of Co_3O_4 phase which reduces in two steps ($Co_3O_4 \rightarrow CoO \rightarrow Co^0$) on the surface of the catalysts [7, 21]. The higher-temperature peaks are identified to the reduction of cobalt oxide(Co_3O_4)-alumina interaction species. The temperature is not so high to observe the complete reduction

Table 2 Surface area, pore volume, average pore size for the catalysts

Catalyst samples	S_{BET} ($m^2\ g^{-1}$)	V_P ($cm^3\ g^{-1}$)[a]	D_P (nm)[b]	d(Co_3O_4) (nm)[c]	d(Co) (nm)[c]
0La–Zr–Co	173	0.26	4.85	18.7	14.3
0.4La–Zr–Co	169	0.26	5.01	18.0	
0.6La–Zr–Co	160	0.22	5.09	16.9	12.4
0.8La–Zr–Co	161	0.23	5.15	17.5	
1.0La–Zr–Co	163	0.24	5.23	17.9	
1.5La–Zr–Co	153	0.25	5.41	18.2	13.6

[a]BJH desorption pore volume

[b]BJH desorption average pore size

[c]Calculated by Scherrer's equation

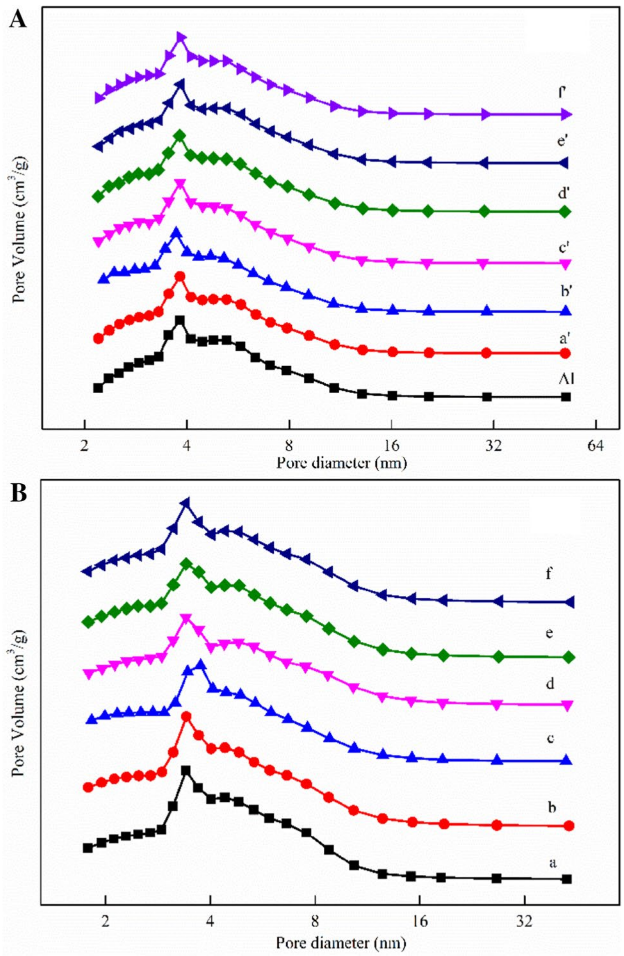

Fig. 1 The pore distribution of the supports (**a**); and the catalysts (**b**); *a'* 0La–Zr; *b'* 0.4La–Zr; *c'* 0.6La–Zr; *d'* 0.8La–Zr; *e'* 1.0La–Zr; *f'* 1.5La–Zr; *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

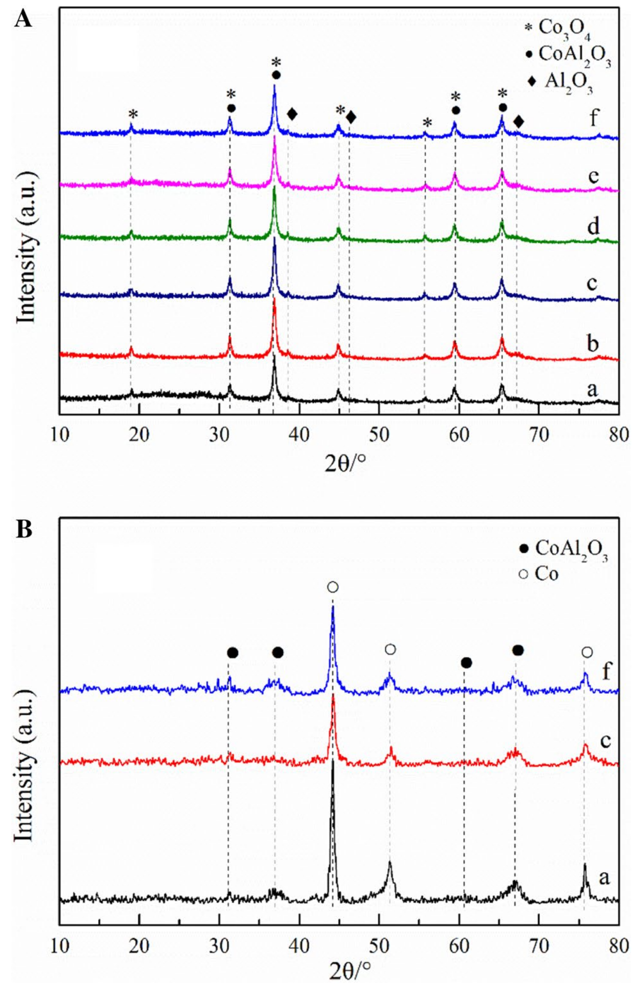

Fig. 2 XRD patterns of the calcined catalysts (**a**), and the reduced catalysts (**b**); *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

of bulk cobalt aluminate (the stoichiometric $CoAl_2O_4$), which have been discussed by Jiao et al. [26] and George et al. [14].

The H_2-TPR results of the catalysts are presented in Table 3. As shown in the Fig. 3 and Table 3, with the addition of La, the first reduction peaks shift to lower reduction temperature and the reduction peaks areas and H_2 consumption increase obviously, while another reduction peaks (500–800 °C) shift slightly to a higher temperature. However, the first reduction peaks shift directionally to the higher temperature with the higher loading of La. It can be reported that with the proper La loading, the degree of reduction of Co_3O_4 to Co^0 is improved, indicating that the interaction between cobalt species with the supports is weakened. Although the H_2 consumption is directly a sign of total available cobalt for the F–T synthesis reaction, the cobalt phase should be of an appropriate size which is suitable for the reaction.

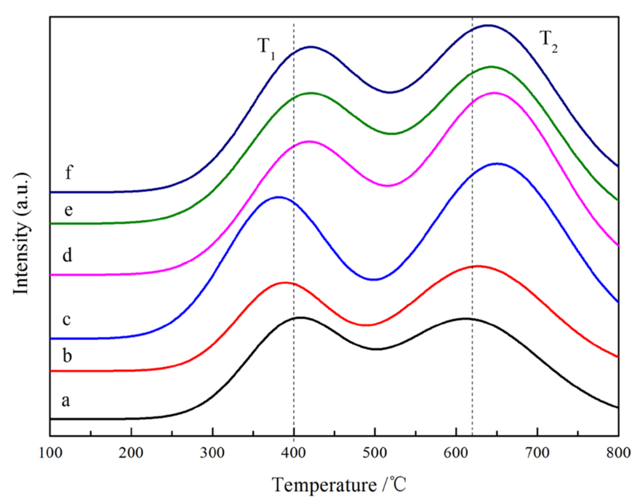

Fig. 3 H_2-TPR profiles of samples with different lanthanum loading. *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

Table 3 The H_2-TPR results of the catalysts

Catalysts	Peak temperature (°C)		H_2 consumption (mmol g^{-1})		Relative reduction degree (%)[a]
	T_1	T_2	T_1	T_2	
0La–Zr–Co	408.05	614.78	1.65	2.30	78
0.4La–Zr–Co	387.60	628.26	1.78	2.38	83
0.6La–Zr–Co	379.27	649.76	2.74	2.78	100
0.8La–Zr–Co	417.43	646.73	1.91	2.43	90
1.0La–Zr–Co	422.44	641.58	1.83	2.37	86
1.5La–Zr–Co	423.50	637.49	1.72	2.32	82

[a]Estimated based on the reduction degree of 0.6La–Zr–Co

Table 4 The CO-TPD results of the catalysts

Catalysts	Peak temperature (°C)		Peak area (a.u.)	
	T_1	T_2	T_1	T_2
0La–Zr–Co	168.44	603.19	0.44	0.18
0.4La–Zr–Co	170.20	602.24	0.34	0.35
0.6La–Zr–Co	174.43	588.83	0.32	0.48
0.8La–Zr–Co	170.10	590.03	0.34	0.31
1.0La–Zr–Co	168.29	600.41	0.37	0.25
1.5La–Zr–Co	168.20	617.61	0.42	0.19

3.4 CO-Temperature Programmed Desorption

CO-TPD is a measurement to explain the effect of La loading on the CO adsorption behavior. The results of the catalysts are exhibited in Fig. 4 and Table 4. There are two major desorption peaks shown in this spectrum. The first low-temperature desorption peak at 200 °C is referred to the desorption of weakly adsorbed CO, which is formed because of the desorption of adsorbed CO on Co^{2+} and can be assigned to the physisorbed CO desorption, while the high-temperature peak at around 600 °C is attributed to the strongly adsorbed CO which can be assigned to the CO species on metallic cobalt atom and can expose accessible surface to reaction [5]. This figure shows that the high-temperature desorption peaks shift to lower temperature and the areas increase, while the low-temperature peaks areas decrease with the increase of La loading. However, once the content of La loading is out of the proper range, the temperatures of strongly adsorbed CO peaks rise and the areas of high-temperature peaks decrease. The result demonstrates that the addition of proper La can enhance the strongly adsorbed CO as well as decrease the weakly adsorbed CO, indicating that there is higher surface carbon monoxide concentration. Thus, the CO-metal bond is reinforced while the C–O bond can be weakened, which can facilitate the CO dissociation and carbon chain growth [5, 27].

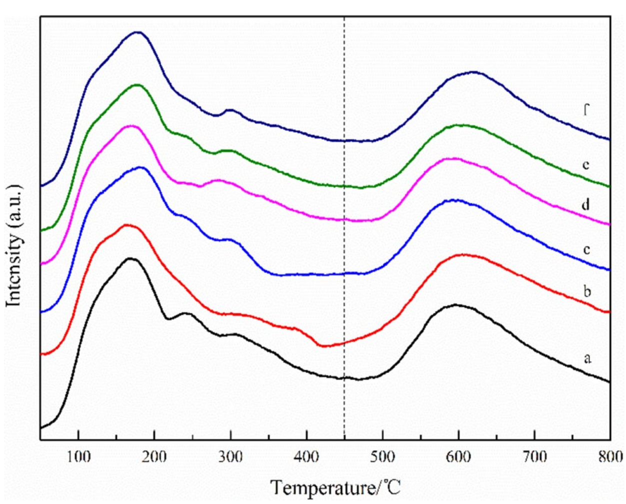

Fig. 4 CO-TPD profiles of samples with different lanthanum loading. *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

3.5 H_2-Temperature Programmed Desorption and H_2-pulse Chemisorption

The H_2-TPD profiles of the catalysts are depicted Fig. 5a, b. Two major hydrogen desorption peaks can be found, which should be explained to two kinds of hydrogen adsorption types on the cobalt-based catalysts. One peak is located from 100 to 200 °C and the other is at approximately 600 °C. Table 5 shows the results calculated from H_2-pulse chemisorption and H_2-TPD experiments. And the overall dispersion of reduced Co in the catalyst samples is basesd on the H_2 chemisorption data and on assumption of H/Co = 1.

As the previous studies reported [28, 29], the lower temperature peak is attributed to the chemisorbed hydrogen on the metal surface, while the higher temperature peak is ascribed to the spillover hydrogen on the support. However, whether the spill-over hydrogen participates in the CO hydrogenation reaction is not clear [30].

It can be revealed that there are changes in the peak position of lower temperature hydrogen desorption with the addition of La loading. The first peaks shift towards to lower temperature and there is an increase in the areas of the lower temperature peaks. From Table 5, the results indicate that with the proper loading of La, the chemisorbed H_2 of reduced catalyst and the dispersion of cobalt metal have an increase, which might be attributed to that more cobalt active sites are exposed on the surface of catalysts [23]. However, with the gradual increase of La content, the lower-temperature peaks shift to a higher temperature and the Co dispersion decreases. The results can be deduced to

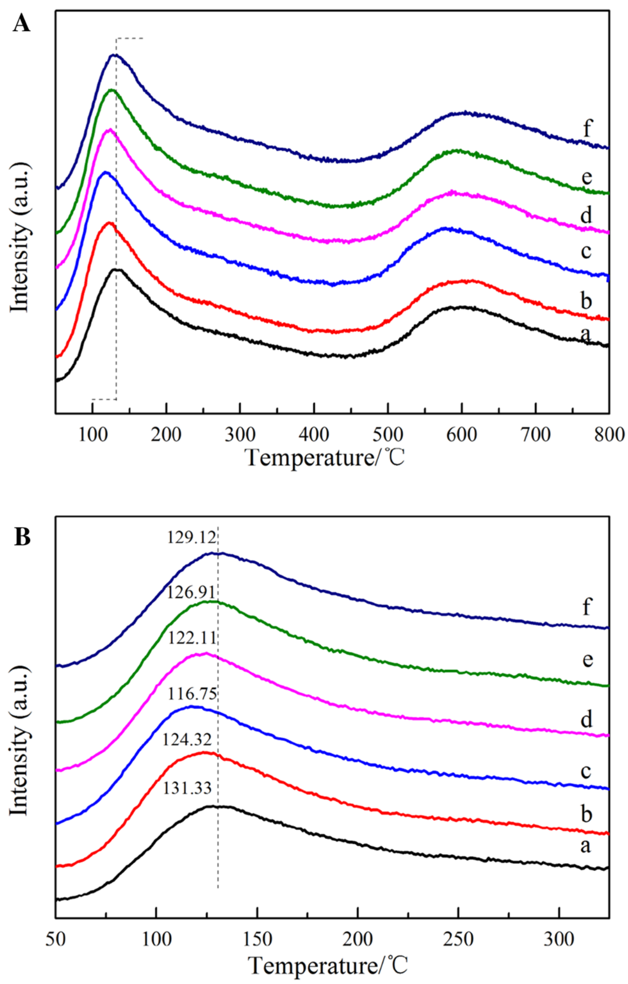

Fig. 5 H_2-TPD profiles of samples (**a**), and enlargement of peaks in lower temperature (**b**). *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

that excessive content of La can block some pores of catalysts, hinder the effects of La on the cobalt dispersion, catalysts reducibility, and cobalt active sites.

3.6 X-Ray Photoelectron Spectroscopy

To investigate the promotion effects of La in the catalysts, the X-ray photoelectron spectroscopy (XPS) is employed to obtain more insights. From the Fig. 6, The Co with different promoter has the binding energies of corresponding to the Co $2p_{3/2}$ component with 781.17, 780.53 eV respectively, which is indicative of the presence of Co_3O_4 and $CoAl_2O_4$, and the literature data of various Co-containing compounds are shown in Table 6 [31]. The binding energy of Co $2p_{3/2}$ is 781.17 eV for 0La–Zr–Co, and shifts toward lower binding energy with the addition of La species. The intensity has no obvious change. It is attributed that La is an electron-donation promoter, can enrich the electron cloud density of Co atoms by electron transfer, and then lead to the decrease of binding energy. Thus, it can reinforce the bond between cobalt and carbon monoxide, facilitating CO dissociation [32], which can be also revealed by CO-TPD. At the same time, it can be revealed that the information of $CoAl_2O_4$ can be inhibited and the interaction between cobalt species and support can be weakened. However, it is observed that the Co $2p_{3/2}$ peak shifts to the higher binding energy for excessive La content, which may be attributed to that high-loading La can result in the coverage of support with layers of lanthanum and the agglomeration of cobalt species, and weaken the electron-donating effect of La [33]. The results are in agreement with H_2-TPR data.

3.7 F–T synthesis with Cobalt-based Catalysts

In order to investigate different microstructures by surface composition effects on the catalytic activity, the F–T synthesis reactions were performed under the conditions of 220 °C, 2.0 MPa, 1000 mL g^{-1} h^{-1} and H_2/CO = 2.0 in a fixed bed reactor. The results of F–T synthesis activity and the product distribution for the catalysts are represented in Figs. 7, 8 and Table 7. CO conversions are plotted as the function of time-on-steam, as presented in Fig. 9. The addition of La to the catalysts has higher CO conversion

Table 5 The H_2-pulse chemisorption and H_2-TPD results of the catalysts

Catalysts	H_2-TPD peak temperature (°C)		Adsorption quantity of H_2 (mmol g^{-1})	Dispersion (%)[a]	dp (Co^0) (nm)[b]
	T_1	T_2			
0La–Zr–Co	131.33	598.62	0.0063	6.5	14.8
0.4La–Zr–Co	124.32	593.43	0.0070	7.1	13.5
0.6La–Zr–Co	116.75	578.02	0.0082	7.9	12.2
0.8La–Zr–Co	122.11	588.24	0.0075	7.3	13.2
1.0La–Zr–Co	126.91	591.65	0.0071	7.2	13.4
1.5La–Zr–Co	129.12	602.03	0.0068	6.8	14.1

[a]Based on the assumption of H/Co = 1

[b]Estimated by Dispersion (D), the corresponding the equation: D = 96/dp (Co^0)

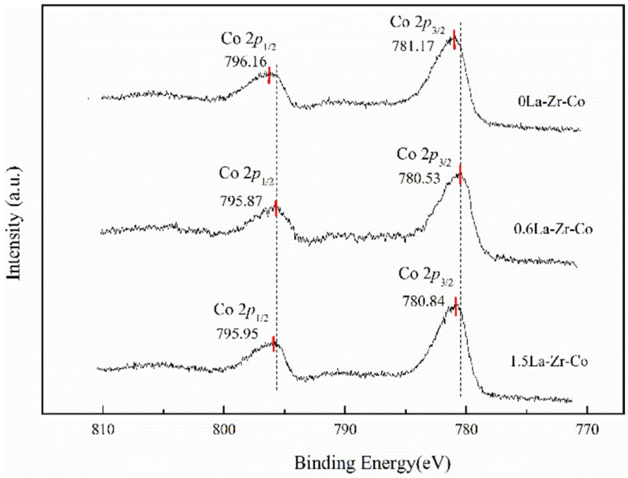

Fig. 6 The XPS spectra of the prepared catalysts

Table 6 The XPS data of various Co-containing compounds

Materials	Co $2p_{3/2}$, BE (eV)	Reliability (eV)
Co_3O_4	780.0	±0.7
$CoAl_2O_4$	781.9	±0.5

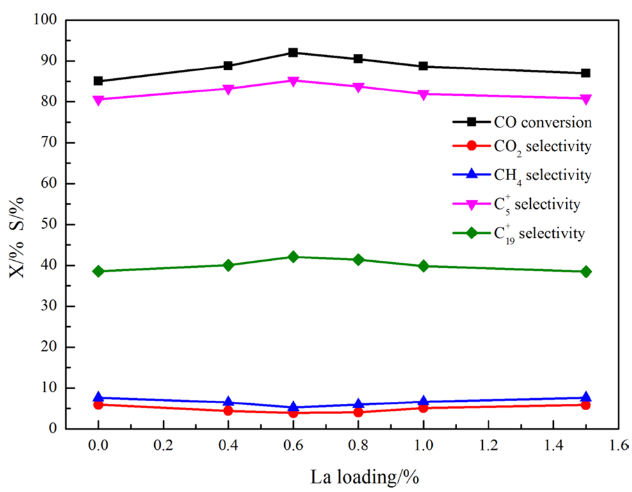

Fig. 7 Catalyst performance of the prepared catalysts for F–T synthesis

and selectivity to C_5^+ and C_{19}^+ hydrocarbon. It could also be summarized that the methane and CO_2 selectivity decrease with the increase of La loading. Moreover, a maximum conversion of 91.97%, the C_5^+ selectivity of 85.23% and C_{19}^+ selectivity of 42.11% are reached over 0.6 wt% La. No obvious deactivation within 60 h can be found among all the catalysts. The observed performance could be obtained based on the characterization results.

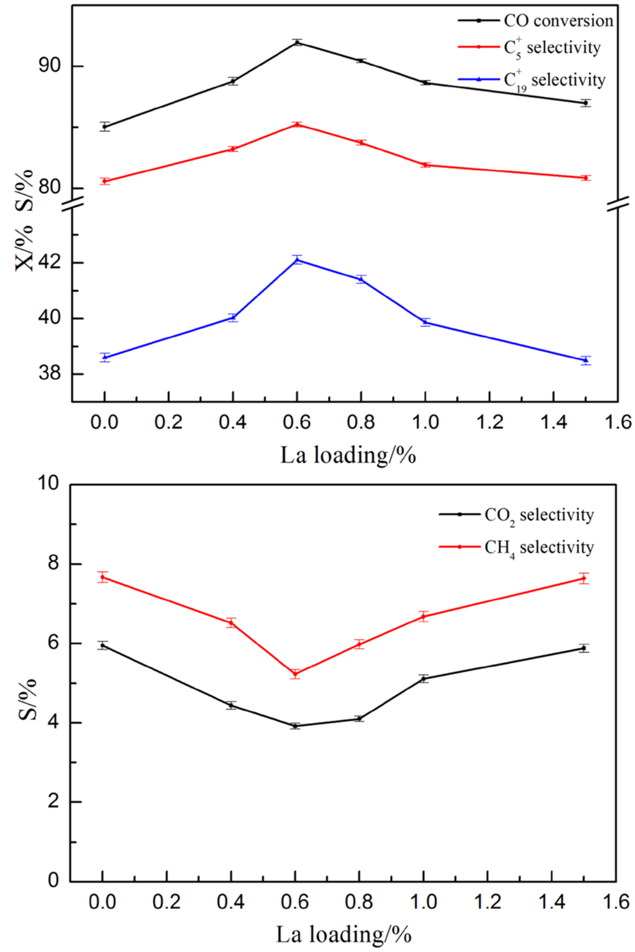

Fig. 8 Catalyst performance with error bars of the prepared catalysts for F–T synthesis

As indicated by H_2-TPD, the proper loading of lanthanum can enhance the cobalt dispersion and make the Co_3O_4 average crystallite size near the optimum size. Well-dispersion of cobalt can improve the cobalt oxide reduction (Co_3O_4 to Co^0), inhibit the information of $CoAl_2O_4$, weaken the interaction between cobalt with support, and then increase more cobalt active sites for catalysts, which is consistent with the results of H_2-TPR and XPS. Meanwhile, by using XPS, it is seen that the electron cloud density of Co atoms also increases with the addition of La. So, the addition of proper La can lead to the increase of cobalt reducibility, active sites and electron density of Co atoms, improving CO conversion [24, 34]. The catalysts selectivity to heavy hydrocarbon can be explained by the H_2/CO ratio on the surface of the catalysts. According to the XPS and CO-TPD data, incorporation of proper La promoter enhances the increase of the high-temperature desorption peaks areas and decrease the binding energy. These results can be deduced to that the increase of electron cloud density of Co atoms can reinforce the bond between cobalt and carbon monoxide [35]. Accordingly,

Table 7 Catalyst performance of the prepared catalysts for F–T synthesis

La (wt%)	Catalysts	CO conv (%)	Selectivity (%)				
			CO_2	CH_4	$C_{2\sim4}$	C_5^+	C_{19}^+
0	0La–Zr–Co	85.05	5.95	7.67	5.81	80.57	38.60
0.4	0.4La–Zr–Co	88.78	4.44	6.52	5.82	83.22	40.03
0.6	0.6La–Zr–Co	91.97	3.92	5.23	5.62	85.23	42.11
0.8	0.8La–Zr–Co	90.46	4.10	5.98	6.17	83.75	41.40
1.0	1.0La–Zr–Co	88.66	5.11	6.77	6.20	81.92	39.86
1.5	1.5La–Zr–Co	86.99	5.81	7.14	6.21	80.84	38.49

Reaction conditions: 220 °C, 2.0 MPa, 1000 mL g^{-1} h^{-1} and H_2/CO = 2.0

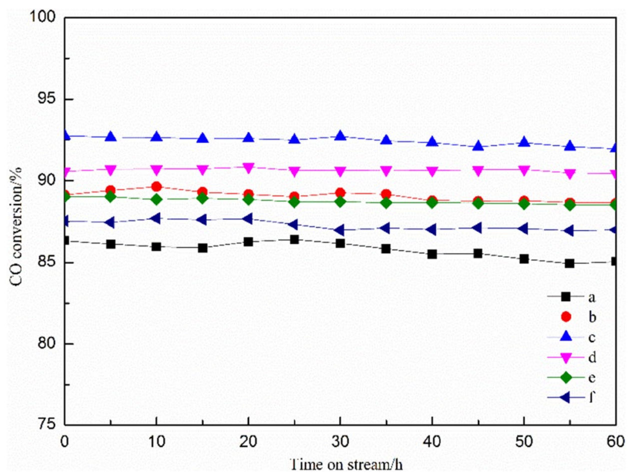

Fig. 9 CO conversion of the catalysts. *a* 0La–Zr–Co; *b* 0.4La–Zr–Co; *c* 0.6La–Zr–Co; *d* 0.8La–Zr–Co; *e* 1.0La–Zr–Co; *f* 1.5La–Zr–Co

carbon monoxide is chemisorbed more strongly and dissociation of CO is easier on the cobalt surface than H_2 adsorption, leading to the possible secondary reaction of olefins through chain growth mechanism and the higher selectivity to heavy hydrocarbon which has been studied in some previous paper [36, 37]. According to some literatures [38, 39], the CO_2 selectivity is relatively high, which might be the reason that partial oxidation of Co at high conversions (high partial pressure of water) could promote WGS (water gas shift) activity of the catalysts during the reaction. Bukur et al. [38] studied that Co metal was reoxidized at high CO conversions (>70%) due to the high water partial pressures, resulting the increases in WGS reaction. And then WGS reaction can lead to a higher H_2/CO ratio on the surface Co^0, which in turn trends to the hydrogenation of adsorbed species to lower carbon number product [40]. Thus, with the addition of La, there are decreases in CH_4 and CO_2 selectivity which can be related to the increase extent of Co reduction. So, higher reducibility of catalysts can result in lower WGS activity, and then lower CH_4 and CO_2 selectivity. However, when the excessive La adds to the catalysts, the BET surface areas become smaller and much more pores of γ-Al_2O_3 support are blocked, resulting in the high coverage of La on support and the agglomeration of cobalt species, and then decreasing the cobalt reducibility and active sites. These reasons lead to lower catalytic activity and C_5^+selectivity and higher methane and CO_2 selectivity after adding higher-loading La.

Thus, the proper content of La promoters with cobalt-based catalysts effectively promotes the overall activity. La promoter can increase the number of active cobalt metal sites and greatly improve the reducibility of catalysts for weakening the interaction of γ-Al_2O_3 support with cobalt oxide and enhancing the dispersion of cobalt, leading to the enhancement of catalysts performance.

4 Conclusion

Lanthanum promoted Zr–Co/γ-Al_2O_3 catalysts, with La loading varying from 0 to 1.5 wt%, were investigated for F–T synthesis. A small amount of La addition can improve the utilization of cobalt species by weakening the interaction between cobalt and support and inhibiting the formation of $CoAl_2O_4$ spinel phase. Thus, the reducibility and cobalt metal active sites for catalysts can be enhanced, leading to relatively higher activity and selectivity of the catalysts and suppressing the formation of methane and CO_2. 0.6La–Zr–Co catalyst shows a better catalytic performance.

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